The exponential excursion-height rate of planar Ising disks

Kaiwen Jin
yui.ui.siki@gmail.com

October 13, 2026

Abstract

We determine the exponential rate at which a topology-preserving planar Ising disk reaches a large area during one excursion from a single square. For addition rate β, deletion rate 1, and every $0 < \beta < 1/\kappa$, the excursion maximum ξ satisfies $\lim_{T\to\infty} \mathbb{P}(\xi \geq T)^{1/T} = \kappa\beta$, where κ is the area growth constant of square-lattice self-avoiding polygons. This proves Conjecture 7.11 of Baryshnikov and Onaran. The lower bound comes from concatenating arbitrary fixed-area disks with one connector square and distributing a unit flow over the resulting growth paths. Cumulative column areas recover each completed prefix, which bounds congestion without requiring disjoint paths within a block.

1 The model and the result

The topology-preserving dynamics of Baryshnikov and Onaran [1] samples finite square-lattice disks by adding and deleting squares while preserving both contractibility and local regularity. Their planar ergodicity theorem identifies the subcritical fugacity range. For rare large excursions, Lemma 7.10 of [1] gives an upper bound on the exponential tail rate, and Conjecture 7.11 asks whether this bound is exact. The question concerns an excursion probability rather than an equilibrium count: many shapes may be reached through the same small set of transitions.

For $z \in \mathbb{Z}^2$, let $Q_z = z + [-1/2, 1/2]^2$. A finite collection C of these squares is a *clump* if its geometric union $|C|$ is contractible and regular. Here regularity means that every $x \in |C|$ is either an interior point or has $(B_\varepsilon(x) \cap |C|) \setminus \{x\}$ contractible for all sufficiently small $\varepsilon > 0$. Write $\#C$ for its number of squares and $o = \{Q_0\}$ for the initial state.

A deletion of $Q \neq Q_0$ is legal when $C \setminus \{Q\}$ is a clump and there is a strong deformation retraction H of $|C|$ onto $|C \setminus \{Q\}|$ with $H(Q \times [0,1]) \subseteq Q$. Addition is the inverse move. Starting at o, let C_t be the continuous-time chain with rate β for each legal addition and rate 1 for each legal deletion. Let ζ be its first return to o after departure, and define

$$\xi = \sup_{0\leq t<\zeta} \#C_t, \qquad p_T = \mathbb{P}(\xi \geq T).$$

Let a_n count square-lattice self-avoiding polygons of area n up to translation, with orientations retained, and let $\kappa = \lim_{n\to\infty} a_n^{1/n}$ be their area growth constant.

Theorem 1. *For every fixed $0 < \beta < 1/\kappa$,*

$$\lim_{\substack{T\to\infty \\ T\in\mathbb{N}}} p_T^{1/T} = \kappa\beta. \tag{1}$$

Moreover, with $\mu = \mathbb{E}\zeta \in (0,\infty)$ and $\eta_T = \mu/p_T$, $\lim_{T\to\infty, T\in\mathbb{N}} \eta_T^{1/T} = 1/(\kappa\beta)$.

Thus the area entropy and the fugacity give the exact exponential scale for large excursions. The assertion is pointwise in β throughout the open subcritical interval; it does not prescribe a prefactor or a uniform rate of convergence in β.

We use two planar facts from [1]. Theorem 5.1 states that every nonsingleton clump has two distinct squares whose deletions leave clumps through strong deformation retractions supported inside the removed squares. This geometric statement imposes no distinguished-square exclusion. The source homotopy condition holds for $t < 1$; closedness and continuity give it at $t = 1$ as well. A clump containing Q_0 can therefore be reduced by legal deletions to o: at each stage one of the two geometrically deletable squares differs from Q_0. Reversing this sequence gives a legal monotone growth path retaining Q_0. Remark 6.3 and equation (4) of [1] identify unrooted planar clumps with the polygons counted by a_n and bound the number of origin-containing size-n clumps by na_n.

Theorem 6.5 of [1] gives nonexplosion and ergodicity in the stated range, and its Definition 7.2 records the resulting positive finite mean return time. The source parameters $p_\downarrow = 1$ and $p_\uparrow = \beta$ give exactly our rates. They are admissible since a straight strip gives $a_n \geq 1$, hence $\kappa \geq 1$ and $\beta < 1$. We use the defining growth limit for κ; no additional enumeration inequality is needed.

2 Excursions and conductance

Fix an integer $T \geq 2$. Retain all origin-containing clumps with at most T squares and all legal edges between them. Regard the size-T states as the terminal set A_T. This is a finite graph, and every nonterminal state has a path to o by the deletion argument above. Give an edge from a size-n state to a size-$(n+1)$ state the conductance

$$c(C, C') = \beta^n. \tag{2}$$

Indeed, the reversible weight $w(C) = \beta^{\#C-1}$ gives this conductance from either direction using the specified rates. At states below size T, the conductance ratios are exactly the actual jump probabilities of the original chain stopped at o or A_T.

Let v be 1 at o, 0 on A_T, and, at other states, the probability of hitting o before A_T. Absorption is almost sure: in the finite stopped graph, each state has a positive-probability path to an absorbing state, and a uniform positive lower bound over these finitely many paths can be applied in repeated bounded blocks of jumps. The first-step equation makes v harmonic off the boundary. Put

$$\mathrm{cap}_T = \sum_{C \sim o} c(o, C)(1 - v(C)).$$

There are exactly four legal exits from o, its four edge-neighbor additions, each of conductance β. Corner-only contacts are not regular. Conditioning on the first actual jump therefore gives

$$p_T = \frac{\mathrm{cap}_T}{4\beta}. \tag{3}$$

Holding times do not affect this event. Finite summation by parts gives

$$\mathrm{cap}_T = \sum_{\{x,y\}} c(x, y)(v(x) - v(y))^2, \tag{4}$$

where undirected edges are counted once: the interior terms vanish by harmonicity, the terminal terms by $v = 0$, and only the term at o remains.

For completeness, the two variational bounds needed below follow directly from this identity. If $f(o) = 1$ and $f = 0$ on A_T, writing $f = v + g$ and summing the cross term by parts gives

$$\sum_{\{x,y\}} c(x,y)(f(x) - f(y))^2 = \mathrm{cap}_T + \sum_{\{x,y\}} c(x,y)(g(x) - g(y))^2 \geq \mathrm{cap}_T \,.$$

Choose $f = 1$ below size T and $f = 0$ at size T. There are at most T downward edges from each of at most Ta_T terminal states, so

$$\mathrm{cap}_T \leq T^2 a_T \beta^{T-1}, \qquad p_T \leq \tfrac{1}{4} T^2 a_T \beta^{T-2}. \tag{5}$$

Taking Tth roots proves $\limsup_{T\to\infty} p_T^{1/T} \leq \kappa\beta$.

For the lower bound, orient each graph edge once. A unit flow F from o to A_T has divergence 1 at o, zero off the boundary, and total divergence -1 on A_T. Its energy is

$$\mathcal{E}(F) = \sum_e \frac{F(e)^2}{c(e)}.$$

Summation by parts and then Cauchy–Schwarz yield

$$1 = \sum_e F(e)(v(\mathrm{tail}\, e) - v(\mathrm{head}\, e)), \qquad 1 \leq \mathcal{E}(F)\,\mathrm{cap}_T \,.$$

Thus any such flow proves $\mathrm{cap}_T \geq 1/\mathcal{E}(F)$.

3 Concatenated disks and the lower bound

We construct many growth paths with controlled congestion by appending arbitrary disks in disjoint columns. Fix an integer $m \geq 1$. Normalize each translation class of m-square clumps so that the lowest square in its leftmost occupied column is Q_0. Denote these representatives by $D_1, \ldots, D_N$, where $N = a_m$, and put $L = m + 1$. For each D_i choose one legal growth path from its port Q_0 to D_i, retaining that port throughout.

Lemma 2. *Let (r, s) be the lowest center in the rightmost occupied column of a clump P containing Q_0. Then*

$$P * D_i = P \cup \{Q_{(r+1,s)}\} \cup (D_i + (r + 2, s))$$

is a clump. It can be reached from P by exactly L legal additions, all strictly to the right of the rightmost column of P.

Proof. Add the bridge $B = Q_{(r+1,s)}$, then the new port $Q_{(r+2,s)}$, and then follow the chosen growth path of the translated block. The bridge meets each component in one full vertical port edge; possible contacts with an above-port square are endpoints of that same edge. No square lies below either port in its extreme column, and the old component and new block have no contact with each other.

At every stage the union is regular. Away from the bridge edges this is the regularity of the individual components. At a lower bridge endpoint the occupied sectors are two adjacent quadrants. At an upper endpoint they are two adjacent quadrants or three consecutive quadrants, according to whether the square above the port is present. Each punctured local neighborhood is contractible; points inside a shared edge become interior.

The partial block retracts to its port by reversing its rooted growth sequence. Each homotopy fixes the port, so it extends by the identity on $P \cup \{B\}$. The port and then the bridge retract horizontally onto their attached edges. Hence the union retracts onto P and is contractible.

These are also legal single-square moves. For an internal block deletion, extend its strong deformation retraction by the identity on the old component and bridge. Any contact of the deleted square with the bridge is in the retained port, so the definitions agree on the intersection. Finite closed-set pasting gives a continuous homotopy, and the homotopy still stays inside the deleted square. The bridge and initial-port moves use the explicit horizontal retractions just described. Every intermediate union and every deletion result is a clump. Since P contains Q_0, we have $r \geq 0$, and every newly added square has first coordinate larger than r. Thus none is Q_0; extending the translated block homotopies by the identity on P retains the actual pinned square. There are $1 + m = L$ additions. □

Starting from o, a completed word of j blocks has size $n_j = 1 + jL$. The following observation permits shared paths within a block while preventing confusion between different completed parents.

Lemma 3. *Different block words give different completed clumps. During stage j, growth paths from different completed parents of size n_j have no common state or upward edge.*

Proof. Every clump in these paths occupies consecutive columns beginning at zero. For such a clump C, define its cumulative column count by

$$S_C(x) = \#\{Q_{(u,v)} \in C : u \leq x\}.$$

This count strictly increases across occupied integer columns. If C lies on a stage-j path after parent P, all squares added in that stage have larger first coordinate than the rightmost column r of P. Thus $S_C(r) = n_j$, and this is the unique occupied column with that count. Restricting C to columns at most r recovers its parent. The lower endpoint of an upward edge therefore identifies its stage and parent as well.

For a completed word, repeat this recovery at counts $n_0, \ldots, n_{j-1}$. Between two successive cuts the first column is the bridge, and the remaining m squares give the normalized block after translation. Thus the entire word is recovered uniquely. □

We now send a unit flow to area $n_k = 1+kL$. Choose k block labels independently and uniformly from $\{1, \ldots, N\}$, and assign each resulting monotone path mass N^{-k}. Summing these path flows gives a unit flow F_k in the size-n_k stopped graph; set it equal to zero on unused edges.

There are N^j distinct stage-j parents, each receiving mass N^{-j}. Within one parent's stage, every edge has flow at most N^{-j} and the sum of edge flows is LN^{-j}, since every continuation has L edges. Therefore its sum of squared edge flows is at most LN^{-2j}. Lemma 3 makes the edge sets for different parents disjoint. This estimate allows different next-block choices within the same parent to share or merge their microscopic growth paths.

The lower size of a stage-j edge lies between $1 + jL$ and $(j + 1)L$. Since $\beta < 1$, equation (2) gives $c(e) \geq \beta^{(j+1)L}$. Consequently

$$\mathcal{E}(F_k) \leq L\beta^{-L} \sum_{j=0}^{k-1} (N\beta^L)^{-j}. \tag{6}$$

Proof of Theorem 1. Set $r_m = a_m \beta^{m+1}$. Its $(m + 1)$st root tends to $\kappa\beta < 1$, so $r_m < 1$ for every sufficiently large m. Fix any such m, and write $r = r_m$. The geometric sum satisfies

$$\sum_{j=0}^{k-1} r^{-j} \leq \frac{r^{-(k-1)}}{1 - r}.$$

The unit-flow bound, equations (3) and (6), give

$$p_{1+kL} \geq \frac{(1-r)\beta^{L-1}}{4L}\, r^{k-1}, \qquad k \geq 1. \tag{7}$$

For any integer $T \geq 2$, take $k = \lceil (T-1)/L \rceil$. Since $1 + kL \geq T$, the decreasing tail satisfies $p_T \geq p_{1+kL}$. The positive prefactor in (7) is fixed, and $(k-1)/T \to 1/L$. Taking Tth roots yields

$$\liminf_{T\to\infty} p_T^{1/T} \geq r_m^{1/(m+1)} = \beta a_m^{1/(m+1)}.$$

This holds for each sufficiently large fixed m. Letting $m \to \infty$ improves the same lower bound to $\kappa\beta$. Together with (5), it proves (1) for the arbitrary fixed admissible β. Finally, $\mu^{1/T} \to 1$ because $0 < \mu < \infty$, so $\eta_T^{1/T} = \mu^{1/T}/p_T^{1/T} \to 1/(\kappa\beta)$. □

AI assistance. OpenAI Codex assisted with argument development, checking cited sources, and preparing the manuscript. The submitted proof is analytic and uses no computational data. Independent mathematical review is requested.

References

[1] Y. Baryshnikov and E. Onaran, *Ising disks: topology preserving Glauber dynamics*, Journal of Applied and Computational Topology **10** (2026), article 16. doi:10.1007/s41468-026-00248-w. Author preprint: arXiv:2410.22611.